

Answer keys for assignment 1

1. $(49.9461 \text{ amu} \times 0.0435) + (51.9405 \text{ amu} \times 0.8379) + (52.9407 \text{ amu} \times 0.0950) + (53.9389 \text{ amu} \times 0.0236) = 51.99 \text{ amu}$
2. a. $4\text{H}_3\text{PO}_3 \rightarrow 3\text{H}_3\text{PO}_4 + \text{PH}_3$
b. $\text{Fe}_2(\text{SO}_4)_3 + 6\text{NH}_3 + 6\text{H}_2\text{O} \rightarrow 2\text{Fe}(\text{OH})_3 + 3(\text{NH}_4)_2\text{SO}_4$
3. $\%C = (12.0\text{g} / 30.0\text{g}) \times 100 = 40.0\%$
 $\%H = ((2 \times 1.01\text{g}) / 30.0\text{g}) \times 100 = 6.73\%$
 $\%O = 100\% - (40.0\% + 6.73\%) = 53.3\%$
4. a.
Number of moles for $\text{CH}_3\text{CHO} = 20.0 \text{ g} / (44.1 \text{ g/mol}) = 0.454 \text{ mol CH}_3\text{CHO}$
Number of moles for $\text{O}_2 = 10.0 \text{ g} / (32.0 \text{ g/mol}) = 0.313 \text{ mol O}_2$
From the reaction, number of moles for CH_3CHO should be equal to twice the number of moles of O_2 . But the number of moles for CH_3CHO is less than twice the number of moles of O_2 . Therefore, CH_3CHO is the limiting reagent.

b.
From the chemical equation the number of moles of $\text{HC}_2\text{H}_3\text{O}_2$ should be equal to the number of moles of CH_3CHO .
Therefore, the amount of acetic acid $= 0.454 \text{ mol} = 0.454 \text{ mol} \times 60.1 \text{ g/mol} = 27.3 \text{ g}$

c.
 $\% \text{yield} = \text{actual yield} / \text{theoretical yield} = (23.8 \text{ g} / 27.3 \text{ g}) \times 100\% = 87.2\%$

d.
From the chemical equation, 1 mol of CH_3CHO will react with 0.5 mol of O_2 .
Therefore, the number of moles O_2 reacts here will be $= \text{number of moles of CH}_3\text{CHO} \times 0.5 = 0.454 \text{ mol} \times 0.5 = 0.227 \text{ mol}$
The amount of unreacted $\text{O}_2 = (0.313 \text{ mol} - 0.227 \text{ mol}) = 0.086 \text{ mol} = 0.086 \text{ mol} \times 32.0 \text{ g/mol} = 2.75 \text{ g}$
5. The complete ionic equation:
 $\text{ZnS(s)} + 2\text{H}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq}) + \text{H}_2\text{S(g)}$

The net ionic equation:
 $\text{ZnS(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2\text{S(g)}$
6. $(\text{Oxidation number of Cl}) + 3 \times (\text{oxidation number of O}) = -1$
Therefore, $(\text{Oxidation number of Cl}) = -1 - 3 \times (-2) = +5$
7. Number of moles of $\text{NaNO}_3 = 0.38 \text{ g} / (85.997 \text{ g/mol}) = 4.5 \times 10^{-3} \text{ moles}$
Molarity $= \text{moles of solute} / \text{volume of solution in liters}$
 $= (4.5 \times 10^{-3} \text{ mol}) / (50.0 \times 10^{-3} \text{ L})$
 $= 0.089 \text{ M}$